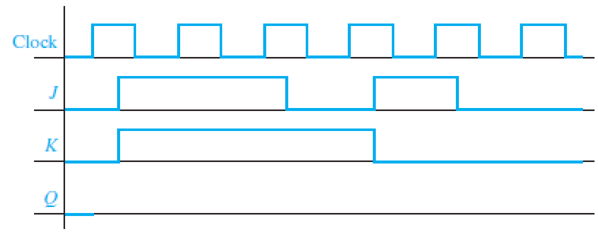
11.7 Complete the following timing diagram for the flip-flop of Figure 11-24(a).

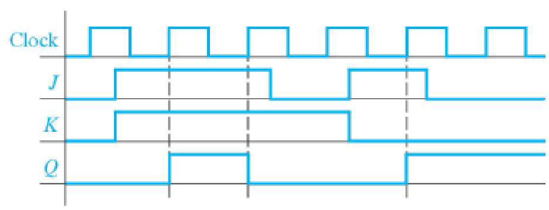


▼ Details

The timing chart for Clock, J, K, and Q. The waveform for Clock is as follows. 0 1 0 1 0 1 0 1 0 1 0. The waveform for J is as follows. 0 1 1 1 1 0 0 1 1 0 0. The waveform for K is as follows. 0 1 1 1 1 1 1 0 0 0 0. The waveform for Q is as follows. 0.

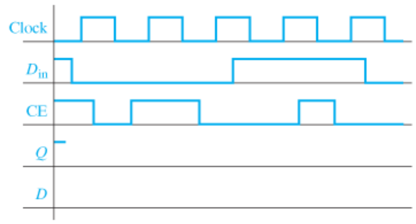
HIDE ANSWER

Answer:



11.8 Complete the following diagrams for the falling-edge-triggered D-CE flip-flop of Figure 11-31(c). Assume Q begins at 1 .

a. First draw Q based on your understanding of the behavior of a D flip-flop with clock enable.



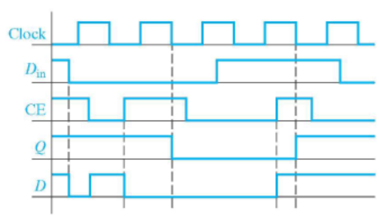
▼ Details

The timing chart for Clock, D_in, C E, Q, and D. The waveform for Clock is as follows. 0 1 0 1 0 1 0 1 0 1 0. The waveform for D_in is as follows. 1 0 0 0 0 1 1 1 1 0 0. The waveform for C E is as follows. 1 1 0 1 1 0 0 0 1 0 0. The waveform for Q is 1. The waveform for D is not provided.

b. Now draw in the internal signal D from Figure 11-31(c), and confirm that this gives the same Q as in (a).

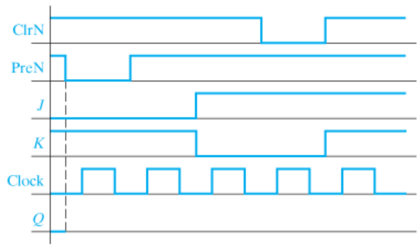
HIDE ANSWER

Answer:



▶ Details

11.9 **a.** Complete the following timing diagram for a J-K flip-flop with a falling-edge trigger and asynchronous ClrN and PreN inputs.



▼ Details

A timing chart for C l r N, Pre N, J, K, Clock, and Q. The waveform for C l r N is as follows. 1 1 1 1 1 1 1 0 0 1 1. The waveform for Pre N is as follows. 1 0 0 1 1 1 1 1 1 1 1. The waveform for J is as follows. 0 0 0 0 0 1 1 1 1 1 1. The waveform for K is as follows. 1 1 1 1 1 0 0 0 0 1 1. The waveform for Clock is as follows. 0 1 0 1 0 1 0 1 0 1 0. The waveform for Q is 0. A vertical dashed line connects the line at Q to the first fall in Pre N.

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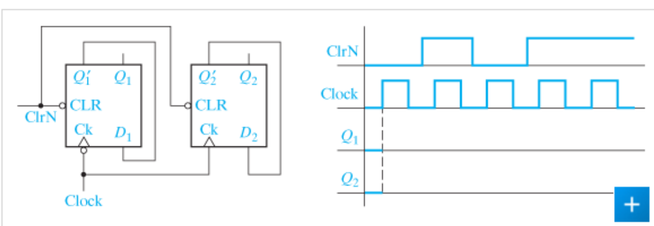
b. Complete the timing diagram for the following circuit. Note that the Ck inputs on the two flip-flops are different.



▶ Details

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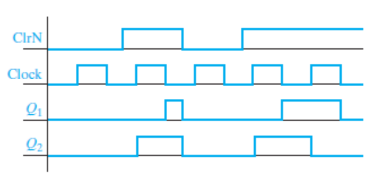
Answer:



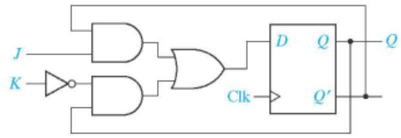
▶ Details

11.10 Convert by adding external gates:

a. a D flip-flop to a J-K flip-flop.

HIDE ANSWER

Answer:

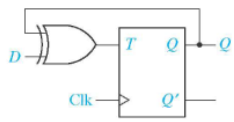


Details

b. a T flip-flop to a D flip-flop.

HIDE ANSWER

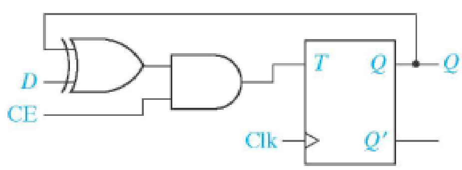
Answer:



c. a T flip-flop to a D flip-flop with clock enable.

HIDE ANSWER

Answer:



Details

Unit 12

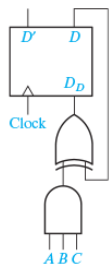
12.4 **a.** Design a 4-bit synchronous binary counter using T flip-flops. (*Hint:* Add one flip-flop, with necessary gates, to the left side of Figure 12-14. Verify that the gates for the other three flip-flops do not change.)

SHOW ANSWER

b. Repeat (a) using D flip-flops. See Figure 12-16.

HIDE ANSWER

Answer:



Details